

A Bilingual Corpus Study of Canadian Algorithmic Impact Assessments

Our next goals:

1. Audit translation consistency and semantic equivalence across the English/French versions of key questions and governance terms (e.g., accountability, fairness, bias/prejudice, responsibility-related language).
2. Map where “untranslatable” or conceptually unstable terms appear, and document how those shifts could plausibly change interpretation and downstream answers.
3. Triangulate: connect linguistic findings to the quantitative patterns we’re seeing (e.g., missingness, shorter answers, different response distributions), so the paper doesn’t just report differences but provides a defensible explanation of mechanisms.
4. Prepare an internal report/draft write-up that consolidates the empirical patterns + linguistic analysis into a coherent narrative suitable for conference presentation and later publication development.

Task 1 : Data Exploration and Confirmation (Augustine)

Try to confirm that the raw data matches what the dashboard reports. This builds your trust in the data pipeline before you tackle the interpretive work.

Task 1.1: Verify the 0% French fields

- Open the file: etl/output/section/about_the_decision.csv
- Look at columns evaluation_criteria_fr and system_output_fr
- Confirm these columns are empty/NULL for all 30 rows
- Record: Are they truly empty, or does any row contain French text?

Task 1.2: Check 5 "Group A" assessments

- Pick 5 assessments that the dashboard (Page 8) classifies as "thorough"
- Open their EN and FR source files in etl/output/extracted/ or data/pdfs/
- Confirm that French fields are actually populated with substantive content
- Record: Does the data match the dashboard's classification?

Task 1.3: Check 5 "Group B" assessments

- Pick 5 assessments classified as "incomplete" on Page 8
- Confirm that French fields are truly empty or minimal
- Record: Does the data match?

Task 1.4: Verify field completion counts

- Open dashboard Page 2 (Inconsistency Problem)
- Cross-reference the completion percentages against the CSV files in etl/output/section/
- Spot-check 3–4 fields: count non-empty rows yourself and compare to dashboard %

Submit: A short note (1 page) confirming or correcting the structural data claims.

Task 2 — Close Reading Report Review (Ralph)

Our goal is to work on the document files here  Reports We are to review their French-relevant claims and rhetorical strategy labels.

Let's break it down

2.1: Review rhetorical strategy labels

- The reports name strategies like "logistical sanitization," "clerical reductionism," "technological diminishment," and "procedural insulation"
- For each label, read the passages cited as evidence
- Question: Are these fair characterizations of what the text is doing, or are they over-interpreting bureaucratic boilerplate?

Task 2.2: Check for English-normative bias

- The reports analyze English-language AIAs exclusively
- Question: Would the same rhetorical patterns hold in the French versions? Or does French administrative language frame automation differently?
- Pick 2 AIAs that have both EN/FR versions and compare the framing in both languages

Submission: a 2 page review noting which labels you endorse, which you question, and whether French-language framing differs from English.

TASK 3 (French Speaking Person - Priscilla?)

Background & Purpose

This project analyzes 30 Canadian federal Algorithmic Impact Assessments (AIAs) published on open.canada.ca. The research examines how these assessments function as governance instruments — and where they fall short.

A significant portion of our findings concern the French-language content of these assessments and how governance terminology translates (or fails to translate) across English and French. These findings were generated using two AI-driven methods:

- Llama 3.3 70B — compared paired EN/FR text fields and classified divergences as omission, reframing, translation error, addition, or terminological shift. It also assigned a semantic fidelity score (1–5) to each assessment.
- GPT-4 Turbo + Python Counter() — extracted governance terms from 16 bilingual PDF pairs and mapped English concepts to their French equivalents (e.g., "accountability" → reddition de comptes / imputabilité / responsabilité).

Instruction: As a bilingual human reviewer, you provide the essential validation layer that no AI can. You are not here to redo the entire analysis — you are here to test whether the AI's judgments hold up when a fluent bilingual reader examines the actual text. Your agreement or disagreement with the AI findings will either strengthen the research or flag claims that need to be revised before publication.

Blind Divergence Classification

You are to independently classify a sample of EN/FR field pairs to test whether the LLM's divergence categories are accurate. Here is a spreadsheet  Bilingual_Verification_Spreadsheet.xlsx which contain 40 EN/FR text pairs drawn from across the corpus. For each pair, you will classify the relationship WITHOUT seeing the LLM's label first.

Step 1: We have prepared a spreadsheet with 40 pairs (columns: `assessment_name`, `field_name`, `english_text`, `french_text`, `your_classification`, `your_notes`). The pairs are sampled to cover all 5 LLM categories proportionally.

Step 2: Read each pair. Classify it using the codebook above. Write a brief note explaining your reasoning.

Step 3: Do NOT look at the LLM classifications until you have completed all 40.

Step 4: After completing your classifications, we will reveal the LLM labels and compute inter-rater agreement (Cohen's kappa).

Step 5: For any disagreements, write a brief explanation of why you classified differently.

Submission: You will submit a completed classification spreadsheet (40 rows) + a 1-page summary of patterns you noticed (e.g., "The LLM consistently miscategorized X as Y").

Governance Terminology in Context

Now, you are to test whether the governance concept mappings (accountability → 3 French terms, *contrôle* vs *monitoring*, *biais* vs *préjugé*) hold up when read in their actual textual context.

Where to Look

Primary files:

- `research/governance_terminology/terminology_comparison.csv` — the 21-row concept mapping with frequencies
- `research/governance_terminology/full_terminology_analysis.json` — full per-pair analysis (3,228 lines)
- `data/pdfs/en/` and `data/pdfs/fr/` — the 16 bilingual PDF pairs (32 documents total)
- `research/hidden_bilingual_pairs.json` — tells you which EN PDF matches which FR PDF
- Dashboard Page 10 (Governance Terminology) — interactive visualization of the findings

Let's break it down

3.1: The Accountability Split

- The claim: English "accountability" (15 occurrences) maps to three distinct French concepts: reddition de comptes (11), imputabilité (4), responsabilité (1)
- Open 5 French PDFs from data/pdfs/fr/
- Search for each of the three French terms
- For each occurrence, note the sentence and surrounding context
- Question to answer: Do these three terms genuinely carry different governance meanings (reporting upward vs. technical attribution vs. legal/moral responsibility)? Or is the distinction artificial / overstated?

3.2: Contrôle vs Monitoring

- The claim: French uses "contrôle" 27 times vs English "monitoring" 5 times, suggesting French expects active management while English expects passive observation
- Read 10 of the 27 "contrôle" passages in the French PDFs
- Classify each as: (a) active management/control, (b) routine checking/verification, (c) ambiguous, (d) other
- Question to answer: Is "contrôle" really signaling a fundamentally different governance philosophy, or is it just the standard French administrative term for "check"?

Task 3.3: Biais vs Préjugé

- The claim: French splits English "bias" (34) into biais (22, technical) and préjugé (9, social/human)
- Read all 9 "préjugé" passages and 10 of the 22 "biais" passages
- Question to answer: Do the contexts support the technical/social distinction? Or are both used interchangeably in practice?

Task 3.4: What Translates Cleanly

- The claim: abstract principles (fairness→équité, transparency→transparence, discrimination→discrimination) translate cleanly while institutional practices do not

- Spot-check 5 passages for each "clean" pair
- Question to answer: Is the "clean" mapping real, or do you see slippage here too?

Submission: You will submit a 2–3 page assessment for each concept mapping: confirmed / partially confirmed / needs revision. Include representative quotes from the PDFs to support your judgment.

HELPFUL RESOURCES

Project Folder Map

The project lives at: <https://github.com/Sagaustus/A-Bilingual-Corpus-Study-of-Canadian-AIA>

Below are the folders and files most relevant to your work.

Folder / File	What's Inside	When You Need It
data/pdfs/en/	16 English AIA PDFs	Task 3 — reading governance terms in context
data/pdfs/fr/	16 French AIA PDFs	Task 3 — reading governance terms in context
research/hidden_bilingual_pairs.json	Defines which EN/FR PDFs are pairs	Task 3 — knowing which docs to compare
research/governance_terminology/	Term counts, CSVs, figures, analysis JSON	Task 3 — verifying term frequencies
research/governance_terminology/terminology_comparison.csv	21-row concept mapping (EN→FR terms with frequencies)	Task 3 — the core claim to verify

research/governance_terminology/		
full_terminology_analysis.json	Full 3,228-line analysis with per-pair details	Task 3 — deep dive into specific pairs
etl/output/section/	12 CSV files with structured form responses	Task 1 — checking field completion
etl/output/section/		
about_the_decision.csv	Contains evaluation_criteria_en/fr, system_output_en/fr	Task 1 — verifying the 0% FR fields
etl/llm_semantic_etl.py		
(lines 506–638)	The LLM prompt used for bilingual divergence analysis	Task 2 — understanding how AI classified divergences
research/Task_2_EVIDENCE.md	SQL queries and results for bilingual findings	All Tasks — tracing any number to its query
Reports/Reports/	6 close reading reports (.docx)	Task 4 — reviewing rhetorical strategy claims
dashboard/pages/	11 Streamlit page files (.py)	Understanding how findings are presented

Dashboard Page Guide [<https://aia.imrag.ca>]

The Streamlit dashboard visualizes the findings. Below is where to look for each claim you are verifying. The dashboard can be launched with: `streamlit run dashboard/Home.py`

Dashboard Page	File	What It Shows (Relevant to Your Review)
Page 2: Inconsistency Problem	<code>dashboard/pages/2_Inconsistency.py</code>	Field completion rates EN vs FR; dead questions (13 fields with zero variation); two-tier completion split. CHECK: Are the 0% FR fields really empty?
Page 4: Bilingual Gap	<code>dashboard/pages/4_Bilingual_Gap.py</code>	The 96.7% divergence rate; omission vs drift breakdown; semantic fidelity scores; bimodal distribution. THIS IS YOUR PRIMARY VERIFICATION TARGET.
Page 8: Two Regimes	<code>dashboard/pages/8_Two_Regimes.py</code>	Group A (thorough) vs Group B (incomplete) split. CHECK: Does bilingual completeness really correlate with overall assessment quality?
Page 9: Language Patterns	<code>dashboard/pages/</code>	

9_Language_Patterns.py	Text volume by department; word frequency patterns; French text is 8–42% longer than English when it exists. CHECK: Is the "longer French" finding real?
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Page 10: Governance Terminology	dashboard/pages/
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10_Governance_Terminology.py	Accountability fracture (3 French terms); contrôle vs monitoring; biais vs préjugé. THIS IS YOUR SECOND PRIMARY TARGET.
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Page 5: Automation Rhetoric	dashboard/pages/
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5_Automation_Rhetoric.py	Justification themes; the confinement gap. USEFUL CONTEXT for understanding how departments describe their systems.
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Page 7: Safeguard Compliance	dashboard/pages/
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7_Safeguard_Compliance.py	Human override claims (100%); bias testing funnel (67% → 3% public). CONTEXT for understanding what the form asks departments to do.
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Things to Verify through close reading

# Finding	How It Was Your Verification Task Generated
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- 1 96.7% of assessments have LLM (Llama 3.3 EN/FR divergence 70B) compared you agree divergence exists?
paired fields
- 2 71% of divergences are LLM classified 256 Independently classify 40 omissions (not mistranslation) divergences into 5 sampled divergences (blind to LLM labels) types
- 3 Semantic fidelity avg = 2.6/5, LLM scored each Score 10 assessments bimodal distribution assessment 1–5 independently, compare to LLM scores
- 4 "accountability" fractures into String matching + Read passages in context — does 3 French concepts GPT-4 semantic the 3-way split hold? grouping
- 5 "contrôle" (27x) vs String matching in Examine 27 "contrôle" instances "monitoring" (5x) = active vs 32 PDFs — is it always "active passive oversight management"?
- 6 "biais" (22) vs "préjugé" (9) = String matching in Read in context — does this technical vs social 32 PDFs distinction hold in AIA usage?